

Package #159

Apply a covalently bonded heparin coating to the luminal and external surfaces of the tunneled hemodialysis catheter during manufacturing

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'catheter thrombosis' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1919-2076). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 10000010: Injury [RESERVOIR 3ML MMT-332A]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Hemostasis at the edge between physiology and cancer.'; 'Fibrinogen in extracellular matrix remodeling: functional switching, source heterogeneity, and biomaterial translation.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Antithrombotic heparin coating on the catheter surface that inhibits fibrin formation and platelet adhesion

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
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Status: ENGINEERING_DEFINITION. Not physically validated.